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Energy-Workload Coupled Migration for Virtual Power Plants With Data Centers Under Customer Directrix Load-Based Demand Response

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***Abstract*—This paper proposes an energy-workload coupled migration optimization strategy for virtual power plants (VPPs) with data centers (DCs) to enhance resource scheduling flexibility and improve aggregate load tracking against the customer directrix load (CDL) trajectory. A coupled migration framework with antisymmetric workload- and energy-transfer matrices is formulated to coordinate cross-regional workload migration and energy sharing among VPPs. By introducing energy sharing as complementary flexibility, the framework expands the feasible set of aggregate load trajectories for CDL tracking and reduces reliance on intensive workload migration, thereby mitigating potential communication-network congestion among geo-distributed DCs. To address uncertainty modeling under limited historical workload data, a credibility-based fuzzy chance-constrained formulation is established for workload-dependent server-capacity and coupled power-balance constraints, and the corresponding deterministic equivalents are analytically derived as tractable affine constraints compatible with the continuous optimization framework. For scalable multi-VPP cooperation, a VPPO-mediated proportional settlement rule is constructed from standalone and grand-coalition costs to avoid exhaustive coalition evaluation, while a consensus-based alternating direction method of multipliers scheme decomposes the coupled scheduling problem into parallel local subproblems and a coordination problem. Simulations based on real data from Google's multiple DCs demonstrate the effectiveness of the proposed method in improving DR curve tracking precision and reducing operational costs.**

***Index Terms*—Virtual power plant, data centers, fuzzy chance constraint, energy sharing, workload migration.**

I. Introduction

With the rapid development of the global digital economy, particularly the widespread adoption of large-scale artificial intelligence model training, the annual electricity consumption of global data centers (DCs) has reached 800 terawatt-hours [1]–[3]. It is projected that global DC electricity consumption will more than double in the next five years, with a compound annual growth rate of 19.5% [4]. This trend not only places significant pressure on energy supply but also presents a crucial challenge to carbon emission reduction targets, leading to extensive attention from both academia and industry on energy optimization for DCs.

The functionality of large-scale DC clusters in power systems extends beyond that of traditional load nodes, exhibiting temporal and spatial coupled-dimensional regulation characteristics [5]. In the temporal domain, these systems achieve power regulation responses within microsecond to millisecond timeframes through refined server load management and dynamic task scheduling [6]. In the spatial domain, leveraging inter-regional computational migration mechanisms among geo-distributed DCs realizes effective virtual power transfer [7]–[9]. This composite network of intertwined energy and information flows enables DCs to exhibit response characteristics markedly distinct from traditional controllable loads [10]–[13], introducing greater flexibility and complexity to power system operations and optimization.

The coupled regulation capability and diversified energy supply structure of DCs are highly compatible with the customer directrix load (CDL) demand response (DR) mechanism [14], [15]. Unlike traditional DR, the CDL mechanism requires precise full-time load tracking rather than simple peak shaving and valley filling. This aligns with three key technical characteristics of DCs: 1) spatial-temporal coupled-dimensional regulation capability provides a wider load adjustment range; 2) categorizable and delayable computational tasks satisfy the temporal flexibility requirements of CDL tracking; and 3) diversified energy systems offer additional regulation degrees of freedom for precise CDL tracking. This technical compatibility makes DCs well suited for implementing CDL-based DR.

Although CDL-based DR can improve regulation accuracy, its strict full-horizon tracking requirement makes it difficult for an individual participant to maintain both response precision and operational flexibility [16], [17]. This challenge is more pronounced at the level of an individual participant, where workload fluctuations and local renewable uncertainty may jointly weaken tracking capability [18]. Cross-regional workload transfer can provide additional flexibility at the technical level. However, CDL-based DR requires not only internal resource adjustment but also a coordinated external response that can be dispatched, measured, and settled against the target trajectory. In this context, the VPP jointly schedules DC computing flexibility, local PV, and BESS resources to form a unified grid-facing power response [19]–[21]. The VPPO

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coordinates the responses of multiple VPPs for aggregate CDL tracking [22], while inter-VPP workload migration and energy sharing provide complementary spatial flexibility and reduce reliance on cross-regional task transfer.

Existing studies have investigated DC participation in power system operation through local scheduling and internal resource management [5], [23]. Cross-regional workload coordination among geo-distributed DCs has also been explored to improve operating efficiency and exploit spatial flexibility [24]. Grid-interactive operation and cooperative DR participation of DCs have further been studied from the perspectives of market interaction and coordinated dispatch [25]. However, most of these studies mainly focus on improving local operating efficiency, reducing electricity cost, or enabling general grid interaction [26], [27]. By contrast, collaborative CDL-based DR for VPPs with geo-distributed DCs requires geographically distributed DCs to be coordinated as aggregated flexibility resources so that the overall response of the VPP can follow a prescribed load trajectory. This feature makes the problem fundamentally different from conventional DC scheduling and general VPP dispatch, and several key technical challenges therefore remain.

First, existing studies on geo-distributed DC coordination mainly exploit workload migration or cross-site scheduling to improve operating efficiency and utilize spatial flexibility [28], while the flexibility provided by local PV and BESS resources is often considered separately [19], [29]–[31]. However, workload migration changes the spatial distribution of computing demand, whereas energy sharing adjusts the local power balance; their complementary effects on aggregate CDL tracking remain insufficiently studied. Excessive reliance on cross-regional task transfer during periods of high task arrivals or large tracking adjustments may also increase communication load and the potential risk of network congestion [32], [33].

Second, existing studies on uncertainty-aware scheduling mainly rely on feasible-region estimation, information-gap decision models, or data-driven robust/stochastic optimization [34]–[36]. Although these methods are useful for general uncertain scheduling problems, their applicability becomes less straightforward in VPP–DC systems when workload data are sparse, evolving, and difficult to characterize probabilistically. In particular, for emerging workloads with limited historical observations, constructing reliable probability models remains challenging.

Third, existing cooperative approaches still face scalability challenges in multi-VPP settings. In coalition-based coordination, benefit allocation methods often rely on repeated coalition evaluation, which becomes computationally burdensome as the number of participants increases [37], [38]. Therefore, besides modeling coupled flexibility, the collaborative scheduling framework should also maintain computational tractability and coordination scalability.

To address the above issues, this paper investigates the collaborative DR of multiple VPPs with geo-distributed DCs under the CDL-based DR mechanism. The objective is to improve regulation flexibility for precise trajectory tracking through the joint coordination of workload migration and energy sharing across VPPs, thereby reducing over-reliance on cross-regional task transfer and alleviating the risk of communication network congestion. The paper also addresses the scheduling challenge caused by workload and photovoltaic (PV) uncertainty under sparse historical data, such as those associated with large-scale artificial intelligence model training, while maintaining computational tractability and coordination scalability in multi-VPP settings.

The main contributions of this paper are summarized as follows:

1) An energy-workload coupled migration framework is proposed for collaborative CDL tracking among geographically distributed DC-based VPPs. Inter-VPP energy sharing complements workload migration by redistributing power responses across sites, providing an alternative to intensive cross-regional task transfers during workload peaks or periods of substantial load adjustment and thus reducing the associated communication-network congestion risk. When workload transfer is constrained, this coupling can expand the feasible set of aggregate load trajectories and improve tracking flexibility.
2) A credibility-based fuzzy chance-constrained formulation is developed to characterize DC workload uncertainty when historical workload data are limited, while also accounting for PV-generation uncertainty. The resulting server-capacity and coupled power-balance chance constraints are transformed into deterministic affine equivalents, while the nonlinear QoS and CDL-tracking terms are reformulated as second-order cone constraints to retain computational tractability. The confidence level controls the trade-off between operating cost and scheduling conservativeness without requiring accurately fitted probability distributions.
3) A VPPO-mediated coordination and settlement mechanism is developed for aggregate CDL tracking among independently managed DC-based VPPs. An ADMM-based consensus scheme separates local scheduling from system-level coordination while maintaining consistency among workload migration, energy sharing, and grid responses. The resulting grand-coalition and standalone costs support a proportional settlement rule that ensures budget balance and individual rationality for a beneficial grand coalition, requiring only $N + 1$ coalition-cost evaluations and avoiding the exponential complexity of the standard Shapley value.

The rest of this paper is organized as follows. Section II presents the system model and problem formulation. Section III introduces the bidirectional migration strategies and uncertainty model construction. Section IV presents the solution methods. Section V reports the simulation results and analysis. Section VI concludes the paper.

II. System Model and Problem Formulation

A. System Description

This paper proposes an energy-workload coupled migration framework to address CDL-based DR challenges in trajectory

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tracking under limited flexibility and multi-source uncertainties. Unlike a conventional peer-to-peer DC interconnection system, the proposed framework is organized under the coordination of a VPP operator (VPPO), which serves as the external interface for CDL participation and DR settlement, while each VPP remains responsible for local resource scheduling. As illustrated in Fig. 1, the framework comprises N geo-distributed VPPs (set $\mathcal{N} = \{1, 2, \ldots, i, \ldots, N\}$) with operations discretized into scheduling periods $\mathcal{T} = \{1, 2, \ldots, t, \ldots, T\}$.

Each VPP$_i$ ($i \in \mathcal{N}$) independently manages its local resources, including the DC load, photovoltaic (PV) generation, and battery energy storage system (BESS). These heterogeneous computing and energy resources are jointly scheduled within each VPP, and their combined effect is reflected in the grid-purchased power $p^{\mathrm{b}}_{i,t}$ at the VPP grid interface. The geo-distribution further enables workload migration and energy sharing among VPPs to exploit spatial complementarity. Accordingly, the VPPO coordinates the grid-facing power responses of multiple VPPs to form the aggregate response used for CDL tracking.

The information layer forms the top tier, where the power grid communicates standardized CDL trajectories ($\mathbf{L}^{\mathrm{CDL}}$) to the VPPO. The CDL mechanism aims to minimize the Euclidean distance $\|\mathbf{L} - \mathbf{L}^{\mathrm{CDL}}\|_2$ between actual and target load shapes, where $\mathbf{L}$ represents the actual load shape. The VPPO acts as a market intermediary, coordinating the operations of multiple independent VPPs while maintaining individual interest equilibrium.

The control layer implements decision-making across the separate VPPs. Each VPP$_i$ control unit manages its own local resources while coordinating with other VPP$_j$ ($j \in \mathcal{N}$) units for energy and workload exchange. Key local decision variables for each VPP$_i$ include server scheduling vectors ($\boldsymbol{s}_i \in \mathbb{Z}^T$), batch workload energy consumption vector ($\boldsymbol{e}^{\mathrm{b}}_i \in \mathbb{R}^T_+$), energy storage operation matrices ($\boldsymbol{q}_i \in \mathbb{R}^{T\times 2}$), and grid interaction vectors ($\boldsymbol{p}^{\mathrm{b}}_i \in \mathbb{R}^T_+$). Inter-VPP coordination occurs through bidirectional transfers where $\Delta\lambda_{ij,t}$ represents interactive workload transferred from VPP$_i$ to VPP$_j$ in time slot t, while $\Delta p_{ij,t}$ denotes corresponding energy transfer. To maintain system-wide resource conservation, these transfer variables satisfy antisymmetric constraints where $\Delta\lambda_{ij,t} + \Delta\lambda_{ji,t} = 0$ and $\Delta p_{ij,t} + \Delta p_{ji,t} = 0$ for all $i, j \in \mathcal{N}$. These constraints ensure zero net transfer across the VPP network.

The resource layer forms the physical infrastructure foundation. In this layer, each VPP's DC power demands are met through a combination of its local PV generation, BESS discharge, and grid purchases when needed. The geographic distribution of these resources allows VPPs to leverage regional complementarities through coordinated operation.

The system operates in four stages: (1) VPPO receives CDL signals and analyzes VPP resource characteristics; (2) each VPP computes its characteristic function $c(\{i\})$; (3) optimization occurs using bidirectional migration mechanisms; and (4) a cooperative settlement rule allocates the coalition benefit among participating VPPs.

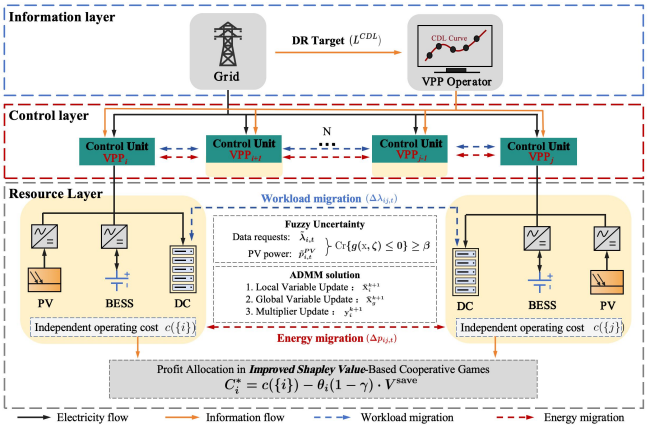


Fig. 1. Framework for geo-distributed DCs participating in DR.

B. Basic Mathematical Model of Each Component

1) DC Model: Batch workloads are scheduled tasks with lower latency sensitivity that can be shifted in the time domain. They use dedicated servers separate from interactive workload servers $\mathbf{s}_i$. In this paper, servers and task arrival rates refer specifically to interactive workloads, while batch workloads are simplified as consuming power $e^b_{i,t}$ in DC$_i$.

This aggregated representation is adopted because the paper focuses on the coordination layer of VPP operation rather than detailed task-level execution within each DC. For delay-tolerant batch jobs, their contribution to inter-temporal flexibility can be effectively captured through aggregate energy demand, which is sufficient for the system-level DR scheduling studied here.

Therefore, DC$_i$'s power consumption in time slot t is:

$$p^{\mathrm{d}}_{i,t} = s_{i,t}\left[e^{\mathrm{idle}} + \left(e^{\mathrm{peak}} - e^{\mathrm{idle}}\right)U_{i,t} + (\eta - 1)e^{\mathrm{peak}}\right] + e^b_{i,t}, \tag{1}$$

$$U_{i,t} = \frac{\lambda_{i,t}}{s_{i,t}u_i}, \tag{2}$$

where $s_{i,t}$ is active server count, $e^{\mathrm{idle}}/e^{\mathrm{peak}}$ are server idle/peak power, η is the power usage effectiveness of the DC, u_i is server processing rate, $\lambda_{i,t}$ is task arrival amount, and $e^b_{i,t}$ is batch workload energy. Denote $\hat{e}^b_i$ as the total energy requirement for batch workloads of VPP$_i$ over the scheduling horizon. The batch workloads must satisfy $\sum_{t\in\mathcal{T}} e^b_{i,t} = \hat{e}^b_i$.

Quality of service (QoS) is ensured through queuing delay cost and server capacity constraints:

$$C^{\mathrm{QoS}}_i = \sum_{t\in\mathcal{T}} \kappa\lambda_{i,t}\left(\frac{1}{s_{i,t}u_i - \lambda_{i,t}}\right), \tag{3}$$

$$\begin{cases} s_{i,t} \le s^{\max}, \\ s_{i,t}u_i \ge \lambda_{i,t}, \end{cases} \quad \forall t, \tag{4}$$

where κ is delay cost coefficient, $s^{\max}$ is maximum server count, and $\frac{1}{s_{i,t}u_i - \lambda_{i,t}}$ is average response time.

2) PV and BESS Models: PV generation and BESS degradation costs are modeled as:

$$C^{\mathrm{PV}}_i = \sum_{t\in\mathcal{T}} \rho^{\mathrm{PV}} p^{\mathrm{PV}}_{i,t}, \tag{5}$$

$$C^{\mathrm{BESS}}_i = \sum_{t\in\mathcal{T}} \varepsilon\left(q^{\mathrm{ch}}_{i,t} + q^{\mathrm{dis}}_{i,t}\right), \tag{6}$$

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$$\mathrm{SoC}_{i,t} = (1-\epsilon)\,\mathrm{SoC}_{i,t-1} + \frac{\eta^{\mathrm{ch}}\, q_{i,t}^{\mathrm{ch}}}{B_i} - \frac{q_{i,t}^{\mathrm{dis}}}{\eta^{\mathrm{dis}}\, B_i}, \tag{7}$$

$$\begin{cases} \mathrm{SoC}_i^{\min} \le \mathrm{SoC}_{i,t} \le \mathrm{SoC}_i^{\max}, & \forall t, \\ 0 \le q_{i,t}^{\mathrm{ch}} \le q_i^{\mathrm{ch,max}}, & \forall t, \\ 0 \le q_{i,t}^{\mathrm{dis}} \le q_i^{\mathrm{dis,max}}, & \forall t, \\ \mathrm{SoC}_{i,0} = \mathrm{SoC}_{i,T}. \end{cases} \tag{8}$$

where ρ^{PV} is the PV unit cost and $p_{i,t}^{\mathrm{PV}}$ is the PV power output. For the BESS, $\varepsilon > 0$ is the degradation coefficient; $q_{i,t}^{\mathrm{ch}}$ and $q_{i,t}^{\mathrm{dis}}$ denote the charging and discharging powers, respectively. $\mathrm{SoC}_{i,t}$ is the state of charge, ϵ is the self-discharge coefficient, B_i is the storage capacity, and η^{ch} and η^{dis} are the charging and discharging efficiencies. The parameters $\mathrm{SoC}_i^{\min}$, $\mathrm{SoC}_i^{\max}$, $q_i^{\mathrm{ch,max}}$, and $q_i^{\mathrm{dis,max}}$ denote the corresponding SoC and power limits. Since the BESS throughput cost is strictly positive, any simultaneous charging and discharging can be reduced while preserving the SoC transition and maintaining the feasibility of the power-balance constraint. Therefore, simultaneous charging and discharging is excluded at optimality without introducing binary mode variables.

C. CDL-based DR Mechanism

The coordinated scheduling of the DC load, local PV generation, and BESS is reflected at the grid interface of VPP_i through the purchased-power trajectory $p_{i,t}^{\mathrm{b}}$, which is linked to the internal resource schedule by

$$p_{i,t}^{\mathrm{b}} \ge p_{i,t}^{\mathrm{d}} - \left(p_{i,t}^{\mathrm{PV}} - q_{i,t}^{\mathrm{ch}} + q_{i,t}^{\mathrm{dis}}\right), \quad \forall t. \tag{9}$$

The grid-interface power response at the point of common coupling (PCC) is subject to the static capacity, the time-varying operating envelope, and the inter-temporal ramping limit:

$$\begin{cases} 0 \le p_{i,t}^{\mathrm{b}} \le P_i^{\mathrm{grid,max}}, & \forall i,t, \\ p_{i,t}^{\mathrm{b}} \le \overline{P}_{i,t}^{\mathrm{OE}}, & \forall i,t, \\ -R_i^{\mathrm{grid}} \le p_{i,t}^{\mathrm{b}} - p_{i,t-1}^{\mathrm{b}} \le R_i^{\mathrm{grid}}, & \forall i,\ t = 2,\dots,T. \end{cases} \tag{10}$$

Here, $P_{i,t}^{\mathrm{OE}}$ is the time-varying PCC import limit derived from local DSO network assessment and treated as an exogenous parameter [39]. Since each modeled VPP has one PCC and only grid imports are considered, a scalar envelope is adopted; multiple coupled connection points would require a generalized operating envelope [40]. The parameters $P_i^{\mathrm{grid,max}}$ and R_i^{grid} denote the static PCC capacity and ramping limit, respectively.

For CDL-based DR, the grid-interface power trajectory $p_{i,t}^{\mathrm{b}}$ is normalized to obtain the response curve $L_{i,t}$ according to (11):

$$\begin{cases} L_{i,t} = \dfrac{p_{i,t}^{\mathrm{b}}}{P_i^D}, \\ d_i = \sqrt{\sum_{t\in\mathcal{T}} \left(L_{i,t} - L_t^{\mathrm{CDL}}\right)^2}, \end{cases} \tag{11}$$

where $\sum_{t\in\mathcal{T}} L_{i,t} = 1$, P_i^D is the declared adjustable load capacity of VPP_i, and L_t^{CDL} is the published CDL. The Euclidean distance d_i measures the deviation between the normalized VPP response and the CDL, with the corresponding similarity defined as $\varepsilon_i = 1 - d_i$. The DR incentive is then calculated as

$$R_i^{\mathrm{DR}} = \rho_o \varepsilon_i P_i^D, \tag{12}$$

where ρ_o is the DR incentive price. The distance-based DR term admits a second-order-cone epigraph representation, which is introduced in the solution method for the independent-operation problem.

III. Bidirectional Migration Strategies and Uncertainty Model Construction

A. Deterministic Bidirectional Migration Model

1) Independent Operation Costs: Under non-cooperative scenarios, the cost minimization problem for independently operating VPPs is formulated as **(P1)**:

$$\min_{\mathbf{X}} \quad C_i - R_i^{\mathrm{DR}}, \tag{13}$$

where $\mathbf{X} = \{s_{i,t}, p_{i,t}^{\mathrm{b}}, q_{i,t}^{\mathrm{ch}}, q_{i,t}^{\mathrm{dis}}, e_{i,t}^{b}\}$ represents the set of decision variables for the independent operations of VPP_i. The total cost function C_i is defined as:

$$C_i = C_i^{\mathrm{PV}} + C_i^{\mathrm{BESS}} + \sum_{t\in\mathcal{T}} \rho_{i,t}^{\mathrm{b}} p_{i,t}^{\mathrm{b}} + C_i^{\mathrm{QoS}}, \tag{14}$$

where $\rho_{i,t}^{\mathrm{b}}$ is the price at which VPP_i purchases electricity from the grid.

2) Coupled Migration Strategy Based on Cooperative Game: The cooperative game comprises players $i \in \mathcal{N}$ representing VPPs participating in DR. VPP_i's strategy set is $\bar{\mathbf{X}} = \{\Delta\lambda_{ij,t}, \Delta p_{ij,t}, \bar{p}_{i,t}^{\mathrm{b}}, s_{i,t}, q_{i,t}^{\mathrm{ch}}, q_{i,t}^{\mathrm{dis}}, e_{i,t}^{b}\}$, where the symbol "$\bar{\cdot}$" represents parameters or decision variables whose mathematical definitions change when workload or energy migration parameters are added. Variables $\Delta\lambda_{ij,t}$ and $\Delta p_{ij,t}$ represent workload migration and energy transfer between VPP_i and VPP_j, respectively.

System cost minimization problem (**P2**) in cooperative operation:

$$\min_{\bar{\mathbf{X}}} \quad \sum_{i\in\mathcal{N}} \bar{C}_i + \sum_{i\in\mathcal{N}} C_i^M - \bar{R}^{\mathrm{DR}}, \tag{15}$$

where

$$\bar{C}_i = C_i^{\mathrm{PV}} + C_i^{\mathrm{BESS}} + \sum_{t\in\mathcal{T}} \rho_{i,t}^{\mathrm{b}} \bar{p}_{i,t}^{\mathrm{b}} + \bar{C}_i^{\mathrm{QoS}}, \tag{16}$$

$$C_i^M = C_i^{\mathrm{W}} + C_i^{\mathrm{E}}, \tag{17}$$

Equation (15) minimizes the total system cost, which consists of the local operating costs and migration costs of all VPPs minus the total DR revenue $\bar{R}^{\mathrm{DR}}$.

To enable cooperation, we define the workload migration matrix $\mathbf{\Lambda_t} = [\Delta\lambda_{ij,t}]_{N\times N}$ and the energy sharing matrix $\mathbf{P}_t = [\Delta p_{ij,t}]_{N\times N}$ for each time slot t. Here, the element $\Delta\lambda_{ij,t}$ represents the amount of interactive workload transferred from VPP_i to VPP_j, while $\Delta p_{ij,t}$ denotes the energy transferred between them. A positive value indicates a transfer from i to j, and a negative value indicates the reverse.

These migration matrices are subject to the following constraints:

$$\begin{cases} \Delta\lambda_{ij,t} + \Delta\lambda_{ji,t} = 0, \\ \Delta p_{ij,t} + \Delta p_{ji,t} = 0, \\ \Delta\lambda_{ii,t} = 0, \Delta p_{ii,t} = 0, \\ |\Delta\lambda_{ij,t}| \le \lambda_{ij}^{\max}, |\Delta p_{ij,t}| \le p^{\max}, \end{cases} \quad \forall i,j,t, \tag{18}$$

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where the first two constraints enforce transfer balance, and the third and fourth exclude self-transfers. The workload-transfer limit $\lambda_{ij}^{\max}$ represents the effective communication capacity between geo-distributed DCs i and j, while $p^{\max}$ limits inter-VPP energy transfer. The former accounts for the finite communication resources available for cross-regional workload migration.

Due to workload migration, the actual computation load, server utilization ratio, DC power consumption, and electricity purchase constraint for VPP_i are adjusted as follows:

$$\begin{cases} \bar{p}_{i,t}^{\mathrm{d}} & = s_{i,t}\Big[e^{\mathrm{idle}} + \big(e^{\mathrm{peak}} - e^{\mathrm{idle}}\big)\bar{U}_{i,t} + (\eta-1)e^{\mathrm{peak}}\Big] \\ & \quad + e_{i,t}^{b}, \\ \bar{U}_{i,t} & = \frac{\bar{\lambda}_{i,t}}{s_{i,t}\,u_i}, \\ \bar{\lambda}_{i,t} & = \lambda_{i,t} - \sum_{j\in\mathcal{N}} \Delta\lambda_{ij,t}, \end{cases} \tag{19}$$

Accordingly, the grid-purchased power of VPP_i is linked to the local resource schedule and bilateral energy transfers by

$$\bar{p}_{i,t}^{\mathrm{b}} \geq \bar{p}_{i,t}^{\mathrm{d}} - \Big(p_{i,t}^{\mathrm{PV}} - q_{i,t}^{\mathrm{ch}} + q_{i,t}^{\mathrm{dis}}\Big) + \sum_{j\in\mathcal{N}} \Delta p_{ij,t}, \quad \forall i,t. \tag{20}$$

Under cooperative operation, each VPP's grid-interface power response remains subject to the static PCC capacity, time-varying operating envelope, and inter-temporal ramping limit:

$$\begin{cases} 0 \leq \bar{p}_{i,t}^{\mathrm{b}} \leq P_i^{\mathrm{grid,max}}, & \forall i,t, \\ \bar{p}_{i,t}^{\mathrm{b}} \leq \overline{P}_{i,t}^{\mathrm{OE}}, & \forall i,t, \\ -R_i^{\mathrm{grid}} \leq \bar{p}_{i,t}^{\mathrm{b}} - \bar{p}_{i,t-1}^{\mathrm{b}} \leq R_i^{\mathrm{grid}}, & \forall i,\ t>1. \end{cases} \tag{21}$$

$$\bar{C}_i^{\mathrm{QoS}} = \sum_{t\in\mathcal{T}} \kappa\bar{\lambda}_{i,t}\left(\frac{1}{s_{i,t}\,u_i - \bar{\lambda}_{i,t}}\right). \tag{22}$$

Let $D = [d_{ij}]_{N\times N}$ denote the inter-VPP distance matrix. The distance-weighted migration costs are defined as

$$C_i^{\mathrm{W}} = \sum_{t\in\mathcal{T}}\sum_{j\in\mathcal{N}} \omega_1 d_{ij}\max(0,\Delta\lambda_{ij,t}), \tag{23}$$

$$C_i^{\mathrm{E}} = \sum_{t\in\mathcal{T}}\sum_{j\in\mathcal{N}} \omega_2 d_{ij}\max(0,\Delta p_{ij,t}). \tag{24}$$

where d_{ij} is the distance between VPP_i and VPP_j, and ω_1 and ω_2 are the unit costs of workload and energy transmission, respectively. For workload migration, distance approximates the aggregate communication overhead and latency effects, consistent with [9], [27], [28]. Explicit bandwidth and latency constraints are omitted for tractability and partially represented by ω_1 and the migration limits in (18).

For CDL-based DR, the grid-interface responses of the participating VPPs are aggregated and compared with the CDL trajectory:

$$\begin{cases} \bar{L}_t = \dfrac{\sum_{i\in\mathcal{N}} \bar{p}_{i,t}^{\mathrm{b}}}{\sum_{i\in\mathcal{N}} P_i^D}, \\ \bar{d} = \sqrt{\sum_{t\in\mathcal{T}} \big(\bar{L}_t - L_t^{\mathrm{CDL}}\big)^2}, \end{cases} \tag{25}$$

where $\bar{p}_{i,t}^{\mathrm{b}}$ represents the grid-interface purchased-power response of VPP_i after local resource scheduling and inter-VPP workload and energy coordination. Accordingly, $\bar{L}_t$ represents the normalized aggregate response of the participating VPPs, and $\bar{d}$ measures its deviation from the CDL trajectory L_t^{CDL}. Defining the similarity as $\varepsilon = 1 - \bar{d}$, the total DR revenue is $\bar{R}^{\mathrm{DR}} = \sum_{i\in\mathcal{N}} \rho_o \varepsilon P_i^D$.

B. Proportional Settlement Rule for Cooperative Operation

A VPPO-mediated settlement rule is introduced for cooperative operation among multiple VPPs, using only the standalone and grand-coalition costs to avoid the exhaustive coalition evaluations required by the standard Shapley value. For any coalition $\mathcal{S} \subseteq \mathcal{N}$, $\bar{\mathbf{X}}_{\mathcal{S}}$ comprises its local scheduling and internal workload- and energy-transfer variables, while $\mathcal{F}(\mathcal{S})$ denotes the corresponding feasible set under the cooperative-operation constraints, with transfers to non-members excluded. The minimum net operating cost of $\mathcal{S}$ is defined as

$$c(\mathcal{S}) = \min_{\bar{\mathbf{x}}_{\mathcal{S}}\in\mathcal{F}(\mathcal{S})} \left[\sum_{i\in\mathcal{S}} \big(\bar{C}_i + C_i^M\big) - \bar{R}_{\mathcal{S}}^{\mathrm{DR}}\right]. \tag{26}$$

Here, $\bar{R}_{\mathcal{S}}^{\mathrm{DR}}$ is calculated from the aggregate grid-purchased power of coalition $\mathcal{S}$ using the CDL-based revenue formulation. The standalone cost $c(\{i\})$ and the grand-coalition cost $c(\mathcal{N})$ are obtained from the independent- and cooperative-operation problems, respectively.

The total cost saving achieved by the grand coalition is:

$$V^{\mathrm{save}} = \sum_{i\in\mathcal{N}} c(\{i\}) - c(\mathcal{N}). \tag{27}$$

The VPPO receives a fraction γ of the cooperative saving as the coordination service fee:

$$V^{\mathrm{O}} = \gamma V^{\mathrm{save}}, \tag{28}$$

where $\gamma \in [0,1)$ denotes the service-fee ratio.

The standard Shapley value is defined as:

$$\phi_i(c) = \sum_{S\subseteq\mathcal{N}\setminus\{i\}} \frac{|S|!(N-|S|-1)!}{N!}[c(S\cup\{i\}) - c(S)]. \tag{29}$$

In contrast, the proportional settlement rule requires only the N standalone costs $c(\{i\})$ and the grand-coalition cost $c(\mathcal{N})$:

$$\phi_i'(c) = c(\{i\}) + \frac{c(\{i\})}{\sum_{j\in\mathcal{N}} c(\{j\})}\left[c(\mathcal{N}) - \sum_{j\in\mathcal{N}} c(\{j\})\right]. \tag{30}$$

Accordingly, the number of required coalition evaluations decreases from exponential in N for the standard Shapley value to $N+1$ for the proportional settlement rule.

The cost saving assigned to VPP_i through cooperation is defined as:

$$V_i^{\mathrm{c}} = c(\{i\}) - \phi_i'(c). \tag{31}$$

The corresponding saving ratio is:

$$\theta_i = \frac{c(\{i\})}{\sum_{j\in\mathcal{N}} c(\{j\})}, \quad \forall i\in\mathcal{N}. \tag{32}$$

For $V^{\mathrm{save}} > 0$, this definition is equivalent to $\theta_i = V_i^{\mathrm{c}}/V^{\mathrm{save}}$.

Finally, the allocated cost for each VPP_i (**P3**) is:

$$C_i^* = c(\{i\}) - \theta_i(1-\gamma)V^{\mathrm{save}}. \tag{33}$$

For $V^{\mathrm{save}} \geq 0$ and $\gamma \in [0,1)$, the settlement satisfies $C_i^* \leq c(\{i\})$, so each VPP obtains a nonnegative saving relative to independent operation. Moreover, $\sum_{i\in\mathcal{N}} C_i^* = c(\mathcal{N}) + V^{\mathrm{O}}$.

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The corresponding derivations of individual rationality and budget balance are provided in Appendix A.

C. Uncertainty Modeling

This study considers uncertainties in interactive workload arrivals and PV generation, as batch workloads are typically planned tasks. For independently operating VPPs, the power balance relationship follows from equations (1), (2), and (9):

$$s_{i,t}\left[e^{\text{idle}}+\frac{\left(e^{\text{peak}}-e^{\text{idle}}\right)\lambda_{i,t}}{s_{i,t}u_i}+(\eta-1)e^{\text{peak}}\right] +e^{b}_{i,t}-\left(p^{\text{PV}}_{i,t}-q^{\text{ch}}_{i,t}+q^{\text{dis}}_{i,t}\right)-p^{\text{b}}_{i,t}\le 0,\quad \forall t. \tag{34}$$

In cooperative mode, according to (19), the power balance relationship is:

$$s_{i,t}\left[e^{\text{idle}}+\frac{\left(e^{\text{peak}}-e^{\text{idle}}\right)\left(\lambda_{i,t}-\sum_{j\in\mathcal{N}}\Delta\lambda_{ij,t}\right)}{s_{i,t}u_i} +(\eta-1)e^{\text{peak}}\right]+e^{b}_{i,t}-\left(p^{\text{PV}}_{i,t}-q^{\text{ch}}_{i,t}+q^{\text{dis}}_{i,t}\right) +\sum_{j\in\mathcal{N}}\Delta p_{ij,t}-\bar{p}^{\text{b}}_{i,t}\le 0,\quad \forall i,t. \tag{35}$$

Given the limited historical data, workload arrivals and PV generation are represented by fuzzy parameters $\tilde{\lambda}i,t$ and $\tilde{p}i,t^{\text{PV}}$. Their non-tilded counterparts serve as nominal estimates for deterministic cost evaluation. Workload uncertainty affects the required server capacity as well as DC power demand, whereas PV uncertainty enters the local power balance. Accordingly, the associated feasibility conditions are formulated as credibility-based chance constraints at a prescribed confidence level $\beta\in[1/2,1]$.

For independent operation, the workload-dependent server-capacity constraint is formulated as

$$\text{Cr}\left\{\tilde{\lambda}_{i,t}-s_{i,t}u_i\le 0\right\}\ge\beta,\quad \forall i,t. \tag{36}$$

Under cooperative operation, workload migration modifies the corresponding server-capacity constraint as

$$\text{Cr}\left\{\tilde{\lambda}_{i,t}-\sum_{j\in\mathcal{N}}\Delta\lambda_{ij,t}-s_{i,t}u_i\le 0\right\}\ge\beta,\quad \forall i,t. \tag{37}$$

Moreover, the post-migration workload is required to remain nonnegative:

$$\text{Cr}\left\{\tilde{\lambda}_{i,t}-\sum_{j\in\mathcal{N}}\Delta\lambda_{ij,t}\ge 0\right\}\ge\beta,\quad \forall i,t. \tag{38}$$

The power balance relationships (34) and (35) are correspondingly formulated as credibility-based chance constraints:

$$\text{Cr}\left\{s_{i,t}\left[e^{\text{idle}}+\frac{\left(e^{\text{peak}}-e^{\text{idle}}\right)\tilde{\lambda}_{i,t}}{s_{i,t}u_i}+(\eta-1)e^{\text{peak}}\right] +e^{b}_{i,t}-\left(\tilde{p}^{\text{PV}}_{i,t}-q^{\text{ch}}_{i,t}+q^{\text{dis}}_{i,t}\right)-p^{\text{b}}_{i,t}\le 0\right\}\ge\beta,\quad \forall i,t, \tag{39}$$

$$\text{Cr}\left\{s_{i,t}\left[e^{\text{idle}}+\frac{\left(e^{\text{peak}}-e^{\text{idle}}\right)\left(\tilde{\lambda}_{i,t}-\sum_{j\in\mathcal{N}}\Delta\lambda_{ij,t}\right)}{s_{i,t}u_i} +(\eta-1)e^{\text{peak}}\right]+e^{b}_{i,t}-\left(\tilde{p}^{\text{PV}}_{i,t}-q^{\text{ch}}_{i,t}+q^{\text{dis}}_{i,t}\right) +\sum_{j\in\mathcal{N}}\Delta p_{ij,t}-\bar{p}^{\text{b}}_{i,t}\le 0\right\}\ge\beta,\quad \forall i,t. \tag{40}$$

A trapezoidal fuzzy variable is represented by a quadruple of crisp numbers (r_1,r_2,r_3,r_4), where $r_1\le r_2\le r_3\le r_4$, with the membership function:

$$\mu(x)=\begin{cases}\dfrac{x-r_1}{r_2-r_1}, & \text{if } r_1\le x\le r_2,\\ 1, & \text{if } r_2\le x\le r_3,\\ \dfrac{x-r_4}{r_3-r_4}, & \text{if } r_3\le x\le r_4,\\ 0, & \text{otherwise.}\end{cases} \tag{41}$$

Theorem 1. *Consider an affine function of trapezoidal fuzzy variables:*

$$g(x,\zeta)=h_0(x)+\sum_{k=1}^{K}h_k(x)\zeta_k,$$

where $\zeta_k=(r_{k1},r_{k2},r_{k3},r_{k4})$ is a trapezoidal fuzzy variable, $k=1,2,\ldots,K$. When $\beta\ge 1/2$, the credibility constraint $Cr\{g(x,\zeta)\le 0\}\ge\beta$ has the deterministic equivalent:

$$h_0(x)+(2-2\beta)\sum_{k=1}^{K}\left[r_{k3}h_k^{+}(x)-r_{k2}h_k^{-}(x)\right] +(2\beta-1)\sum_{k=1}^{K}\left[r_{k4}h_k^{+}(x)-r_{k1}h_k^{-}(x)\right]\le 0,$$

where

$$\begin{cases}h_k^{+}(x)=\max\{h_k(x),0\},\\ h_k^{-}(x)=\max\{-h_k(x),0\}.\end{cases}$$

The triangular fuzzy form is adopted because it only requires lower, nominal, and upper estimates, which is suitable when historical data are limited and accurate probability information is difficult to obtain. Assume that the fuzzy parameters $\tilde{\lambda}_{i,t}$ and $\tilde{p}^{\text{PV}}_{i,t}$ follow triangular fuzzy distributions:

$$\begin{cases}\tilde{\lambda}_{i,t}=(a^{\lambda}_{i,t},b^{\lambda}_{i,t},c^{\lambda}_{i,t}),\\ \tilde{p}^{\text{PV}}_{i,t}=(a^{p}_{i,t},b^{p}_{i,t},c^{p}_{i,t}),\end{cases} \tag{42}$$

where a triangular fuzzy distribution is a special case of a trapezoidal fuzzy distribution with $r_1=a$, $r_2=r_3=b$, and $r_4=c$.

According to Theorem 1, the coefficient of $\tilde{\lambda}_{i,t}$ in the power-balance constraints is positive, whereas that of $\tilde{p}^{\text{PV}}_{i,t}$ is negative. Hence, their deterministic equivalents use the upper credibility bound $(2-2\beta)b^{\lambda}_{i,t}+(2\beta-1)c^{\lambda}_{i,t}$ for workload arrivals and the lower credibility bound $(2-2\beta)b^{p}_{i,t}+(2\beta-1)a^{p}_{i,t}$ for PV generation.

From Theorem 1, the deterministic equivalent constraint for

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independent operation is:

$$\begin{aligned} & s_{i,t}\left[e^{\mathrm{idle}}+(\eta-1)e^{\mathrm{peak}}\right]+(q^{\mathrm{ch}}_{i,t}-q^{\mathrm{dis}}_{i,t})-p^{\mathrm{b}}_{i,t} \\ & +\frac{e^{\mathrm{peak}}-e^{\mathrm{idle}}}{u_i}\left[(2-2\beta)b^{\lambda}_{i,t}+(2\beta-1)c^{\lambda}_{i,t}\right]+e^{b}_{i,t} \\ & -\left[(2-2\beta)b^{p}_{i,t}+(2\beta-1)a^{p}_{i,t}\right]\le 0, \quad \forall i,t. \end{aligned} \tag{43}$$

The fuzzy server-capacity constraint (36) has the deterministic equivalent:

$$s_{i,t}u_i \ge (2-2\beta)b^{\lambda}_{i,t}+(2\beta-1)c^{\lambda}_{i,t}, \quad \forall i,t. \tag{44}$$

For consistency with the prescribed confidence level, the workload-dependent utilization and QoS terms are evaluated using the same upper credibility bound:

$$\begin{cases} U_{i,t}=\dfrac{(2-2\beta)b^{\lambda}_{i,t}+(2\beta-1)c^{\lambda}_{i,t}}{s_{i,t}u_i}, \\ C^{\mathrm{QoS}}_{i}=\displaystyle\sum_{t\in\mathcal{T}}\kappa\lambda_{i,t}\left(\frac{1}{s_{i,t}u_i-\left[(2-2\beta)b^{\lambda}_{i,t}+(2\beta-1)c^{\lambda}_{i,t}\right]}\right). \end{cases} \tag{45}$$

The problem **P1** can be expressed as:

$$\begin{cases} \min\limits_{\mathbf{X}} \quad C_i-R^{\mathrm{DR}}_{i} \\ \text{s.t.} \quad (7),(8),(10),(43),(44),(45), \\ \sum_{t\in\mathcal{T}}e^{b}_{i,t}=\hat{e}^{b}_{i}, s_{i,t}\le s^{\max}, \quad \forall t, \\ \beta\ge 1/2, \end{cases} \tag{46}$$

where $\mathbf{X}=\{s_{i,t},p^{\mathrm{b}}_{i,t},q^{\mathrm{ch}}_{i,t},q^{\mathrm{dis}}_{i,t},e^{b}_{i,t}\}$ represents the set of decision variables for the independent operations of VPP$_i$.

The deterministic equivalent constraint for the cooperative case is:

$$\begin{aligned} & s_{i,t}\left[e^{\mathrm{idle}}+(\eta-1)e^{\mathrm{peak}}\right]+(q^{\mathrm{ch}}_{i,t}-q^{\mathrm{dis}}_{i,t}) \\ & +\sum_{j\in\mathcal{N}}\Delta p_{ij,t}-\bar{p}^{\mathrm{b}}_{i,t}-\frac{e^{\mathrm{peak}}-e^{\mathrm{idle}}}{u_i}\sum_{j\in\mathcal{N}}\Delta\lambda_{ij,t} \\ & +\frac{e^{\mathrm{peak}}-e^{\mathrm{idle}}}{u_i}\left[(2-2\beta)b^{\lambda}_{i,t}+(2\beta-1)c^{\lambda}_{i,t}\right] \\ & +e^{b}_{i,t}-\left[(2-2\beta)b^{p}_{i,t}+(2\beta-1)a^{p}_{i,t}\right]\le 0, \quad \forall i,t. \end{aligned} \tag{47}$$

For cooperative operation, the fuzzy server-capacity constraint (37) and post-migration workload constraint (38) are equivalently reformulated as

$$\begin{cases} \bar{\lambda}^{\beta}_{i,t}=(2-2\beta)b^{\lambda}_{i,t}+(2\beta-1)c^{\lambda}_{i,t}-\displaystyle\sum_{j\in\mathcal{N}}\Delta\lambda_{ij,t}, \\ s_{i,t}u_i\ge\bar{\lambda}^{\beta}_{i,t}, \\ \displaystyle\sum_{j\in\mathcal{N}}\Delta\lambda_{ij,t}\le(2-2\beta)b^{\lambda}_{i,t}+(2\beta-1)a^{\lambda}_{i,t}. \end{cases} \quad \forall i,t. \tag{48}$$

Here, $\bar{\lambda}_{i,t}$ remains the nominal post-migration workload defined in (19), whereas $\bar{\lambda}^{\beta}_{i,t}$ is its confidence-adjusted upper bound. As in the independent-operation model, the nominal workload is used to evaluate the delay cost, while the confidence-adjusted value determines the service-capacity margin. The cooperative QoS cost is therefore represented as

$$\bar{C}^{\mathrm{QoS}}_{i}=\sum_{t\in\mathcal{T}}\kappa\bar{\lambda}_{i,t}\left(\frac{1}{s_{i,t}u_i-\bar{\lambda}^{\beta}_{i,t}}\right). \tag{49}$$

The resulting cooperative problem **P2** is formulated as

$$\begin{aligned} \min_{\bar{\mathbf{X}}} \quad & \sum_{i\in\mathcal{N}}\bar{C}_i+\sum_{i\in\mathcal{N}}C^{M}_{i}-\bar{R}^{\mathrm{DR}} \\ \text{s.t.} \quad & (7),(8),(21),(18), \\ & (47),(48),(49), \\ & \sum_{t\in\mathcal{T}}e^{b}_{i,t}=\hat{e}^{b}_{i}, \quad \forall i, \\ & s_{i,t}\le s^{\max}, \quad s_{i,t}\in\mathbb{Z}_{+}, \quad \forall i,t. \end{aligned} \tag{50}$$

where $\bar{\mathbf{X}}=\{\Delta\lambda_{ij,t},\Delta p_{ij,t},\bar{p}^{\mathrm{b}}_{i,t},s_{i,t},q^{\mathrm{ch}}_{i,t},q^{\mathrm{dis}}_{i,t},e^{b}_{i,t}\}$ collects the cooperative decision variables.

D. Feasible-Set Relation of Coupled Migration

Let $\mathcal{F}^{\mathrm{W}}$ and $\mathcal{F}^{\mathrm{EW}}$ denote the feasible sets of aggregate load trajectories under workload-only and energy-workload coupled migration, respectively. Setting $\Delta p_{ij,t}=0$ in (50) recovers the workload-only model without altering the remaining constraints.

$$\mathcal{F}^{\mathrm{W}}\subseteq\mathcal{F}^{\mathrm{EW}}. \tag{51}$$

The inclusion becomes strict when feasible energy sharing alleviates a local energy-supply or grid-interface bottleneck and enables an aggregate trajectory unattainable within the workload-transfer limits.

$$\mathcal{F}^{\mathrm{W}}\subsetneq\mathcal{F}^{\mathrm{EW}}. \tag{52}$$

IV. Solution Methods

A. Solution Method for Independent Operation Scenario

The independent-operation problem (13) contains integer server-count variables, nonlinear QoS and DR terms, and deterministic equivalents of the fuzzy chance constraints.

1) Continuous Relaxation and Integer Feasibility Recovery: The server-count variables $s_{i,t}$ are first relaxed as continuous variables. After solving the relaxed problem, candidate integer values are obtained by upward rounding:

$$\hat{s}_{i,t}=\left\lceil s^{*}_{i,t}\right\rceil, \quad \forall i,t. \tag{53}$$

Since $s^{\max}$ is an integer and the continuous solution satisfies $s^{*}_{i,t}\le s^{\max}$, the rounded values satisfy

$$0\le\hat{s}_{i,t}-s^{*}_{i,t}<1, \quad \forall i,t, \tag{54}$$

$$\hat{s}_{i,t}\le s^{\max}, \quad \forall i,t. \tag{55}$$

Since $u_i>0$ and $\hat{s}_{i,t}\ge s^{*}_{i,t}$, upward rounding preserves the server-capacity constraints in (44) and (48). After fixing $s_{i,t}=\hat{s}_{i,t}$, the remaining continuous variables are re-optimized under the original constraints. Only solutions satisfying all original constraints are retained, and their objective values are evaluated directly without a first-order Taylor approximation.

2) Nonlinear Term Handling: The queuing-delay term in (3) is nonlinear. Define the confidence-adjusted workload bound as

$$\lambda^{\beta}_{i,t}=(2-2\beta)b^{\lambda}_{i,t}+(2\beta-1)c^{\lambda}_{i,t}. \tag{56}$$

Here, $\lambda_{i,t}$ is the nominal workload used to evaluate the QoS cost, whereas $\lambda^{\beta}_{i,t}$ determines the server-capacity margin. Introduce an auxiliary variable $z_{i,t}$ satisfying

$$z_{i,t}\ge\frac{1}{s_{i,t}u_i-\lambda^{\beta}_{i,t}}, \quad \forall i,t, \tag{57}$$

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and represent the conservative QoS cost as

$$C_i^{\mathrm{QoS}} = \sum_{t\in\mathcal{T}} \kappa\lambda_{i,t} z_{i,t}. \tag{58}$$

Because the coefficient of $z_{i,t}$ in the objective is nonnegative, (57) is tight at an optimum whenever $\lambda_{i,t} > 0$. On the feasible domain $s_{i,t}u_i - \lambda^{\beta}_{i,t} > 0$, it can be equivalently written as

$$z_{i,t}\left(s_{i,t}u_i - \lambda^{\beta}_{i,t}\right) \geq 1, \quad \forall i,t, \tag{59}$$

with the following second-order cone representation:

$$\left\| \begin{pmatrix} 2 \\ z_{i,t} - \left(s_{i,t}u_i - \lambda^{\beta}_{i,t}\right) \end{pmatrix} \right\|_2 \leq z_{i,t} + s_{i,t}u_i - \lambda^{\beta}_{i,t}, \quad \forall i,t. \tag{60}$$

For the DR revenue term, define $\mathbf{p}_i^{\mathrm{b}} := [p_{i,1}^{\mathrm{b}}, p_{i,2}^{\mathrm{b}}, \ldots, p_{i,T}^{\mathrm{b}}]$ and $\mathbf{L}^{\mathrm{CDL}} := [L_1^{\mathrm{CDL}}, L_2^{\mathrm{CDL}}, \ldots, L_T^{\mathrm{CDL}}]$. The distance relation in (11) has the epigraph form

$$\left\| \frac{1}{P_i^D}\mathbf{p}_i^{\mathrm{b}} - \mathbf{L}^{\mathrm{CDL}} \right\|_2 \leq d_i. \tag{61}$$

Since the objective $C_i - R_i^{\mathrm{DR}}$ is increasing in d_i, (61) is tight at an optimum. Together with the deterministic chance-constraint equivalents and the convex BESS constraints in (8), the continuous relaxation of **P1** is an SOCP. The integer server counts are then recovered using (53), followed by re-optimization and feasibility verification of the remaining variables.

B. ADMM Decomposition Solution Method under Cooperative Operation

1) Distributed Solution Framework: To solve the continuous relaxation of (50), an ADMM-based distributed framework is adopted to separate local VPP scheduling from system-level coordination. For each VPP i, the local variables are defined as

$$\bar{\boldsymbol{X}}_i = \left\{ s_{i,t}, q_{i,t}^{\mathrm{ch}}, q_{i,t}^{\mathrm{dis}}, e_{i,t}^{b}, \Delta\lambda_{ij,t}^{i}, \Delta p_{ij,t}^{i}, \bar{p}_{i,t}^{\mathrm{b},i} \right\},$$

while the global coordination variables are

$$\bar{\boldsymbol{X}}_g = \left\{ \Delta\lambda_{ij,t}^{g}, \Delta p_{ij,t}^{g}, \bar{p}_{i,t}^{\mathrm{b},g} \right\}.$$

The migration variables are defined for $i \neq j$, while their diagonal elements are fixed at zero. Local and global copies satisfy

$$\begin{aligned} \Delta\lambda_{ij,t}^{i} &= \Delta\lambda_{ij,t}^{g}, && \forall i \neq j, t, \\ \Delta p_{ij,t}^{i} &= \Delta p_{ij,t}^{g}, && \forall i \neq j, t, \\ \bar{p}_{i,t}^{\mathrm{b},i} &= \bar{p}_{i,t}^{\mathrm{b},g}, && \forall i, t. \end{aligned} \tag{62}$$

Introducing d for the aggregate CDL distance yields the following SOC representation of the cooperative DR revenue:

$$\bar{R}^{\mathrm{DR}} = \sum_{i\in\mathcal{N}} \rho_o(1-d)P_i^D, \tag{63}$$

$$\left\| \bar{\boldsymbol{L}} - \boldsymbol{L}^{\mathrm{CDL}} \right\|_2 \leq d. \tag{64}$$

Here, $\bar{\boldsymbol{L}}$ is obtained from $\bar{p}_{i,t}^{\mathrm{b},g}$, while $\boldsymbol{L}^{\mathrm{CDL}}$ denotes the target curve. For $\rho_o > 0$, (64) is tight at an optimum because the objective minimizes $-\bar{R}^{\mathrm{DR}}$. The migration-dependent QoS term is retained in each local objective. Dualizing the consistency constraints in (62), while retaining all other constraints in their respective local or global feasible domains, yields the augmented Lagrangian:

$$\begin{aligned} \mathcal{L}_\rho\big(\{\bar{\boldsymbol{X}}_i\}_{i\in\mathcal{N}}, \bar{\boldsymbol{X}}_g, d, \boldsymbol{y}\big) &= \sum_{i\in\mathcal{N}} \left[\bar{C}_i(\bar{\boldsymbol{X}}_i) + C_i^M(\bar{\boldsymbol{X}}_i)\right] \\ &- \sum_{i\in\mathcal{N}} \rho_o(1-d)P_i^D \\ &+ \sum_{i,j,t} y_{ij,t}^{\lambda}\left(\Delta\lambda_{ij,t}^{i} - \Delta\lambda_{ij,t}^{g}\right) \\ &+ \sum_{i,j,t} y_{ij,t}^{p}\left(\Delta p_{ij,t}^{i} - \Delta p_{ij,t}^{g}\right) \\ &+ \sum_{i,t} y_{i,t}^{b}\left(\bar{p}_{i,t}^{\mathrm{b},i} - \bar{p}_{i,t}^{\mathrm{b},g}\right) \\ &+ \frac{\rho}{2}\sum_{i,j,t}\left(\Delta\lambda_{ij,t}^{i} - \Delta\lambda_{ij,t}^{g}\right)^2 \\ &+ \frac{\rho}{2}\sum_{i,j,t}\left(\Delta p_{ij,t}^{i} - \Delta p_{ij,t}^{g}\right)^2 \\ &+ \frac{\rho}{2}\sum_{i,t}\left(\bar{p}_{i,t}^{\mathrm{b},i} - \bar{p}_{i,t}^{\mathrm{b},g}\right)^2, \end{aligned}$$

where $y_{ij,t}^{\lambda}$, $y_{ij,t}^{p}$, and $y_{i,t}^{b}$ are the Lagrangian multipliers associated with (62), and $\rho > 0$ is the penalty parameter.

At iteration k, each VPP first solves its local subproblem in parallel:

$$\begin{aligned} \bar{\boldsymbol{X}}_i^{k+1} = \arg\min_{\bar{\boldsymbol{X}}_i} \Bigg\{ & \bar{C}_i(\bar{\boldsymbol{X}}_i) + C_i^M(\bar{\boldsymbol{X}}_i) \\ &+ \sum_{j,t} y_{ij,t}^{\lambda,k}\left(\Delta\lambda_{ij,t}^{i} - \Delta\lambda_{ij,t}^{g,k}\right) \\ &+ \sum_{j,t} y_{ij,t}^{p,k}\left(\Delta p_{ij,t}^{i} - \Delta p_{ij,t}^{g,k}\right) \\ &+ \sum_{t} y_{i,t}^{b,k}\left(\bar{p}_{i,t}^{\mathrm{b},i} - \bar{p}_{i,t}^{\mathrm{b},g,k}\right) \\ &+ \frac{\rho}{2}\sum_{j,t}\left(\Delta\lambda_{ij,t}^{i} - \Delta\lambda_{ij,t}^{g,k}\right)^2 \\ &+ \frac{\rho}{2}\sum_{j,t}\left(\Delta p_{ij,t}^{i} - \Delta p_{ij,t}^{g,k}\right)^2 \\ &+ \frac{\rho}{2}\sum_{t}\left(\bar{p}_{i,t}^{\mathrm{b},i} - \bar{p}_{i,t}^{\mathrm{b},g,k}\right)^2 \Bigg\}. \end{aligned}$$

The local subproblem is subject to (7), (8), (21), (47), and (48), together with

$$\begin{aligned} \sum_{t\in\mathcal{T}} e_{i,t}^{b} &= \hat{e}_i^{b}, \\ 0 \leq s_{i,t} &\leq s^{\max}, \quad \forall t, \\ \bar{\lambda}_{i,t} &\geq 0, \quad \forall t. \end{aligned}$$

Second, the global migration variables are updated for each

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unordered pair $i<j$ as

$$\Delta\lambda_{ij,t}^{g,k+1}=\Pi_{\lambda_{ij}^{\max}}\left(\frac{\Delta\lambda_{ij,t}^{i,k+1}-\Delta\lambda_{ji,t}^{j,k+1}}{2}+\frac{y_{ij,t}^{\lambda,k}-y_{ji,t}^{\lambda,k}}{2\rho}\right),$$
$$\Delta\lambda_{ji,t}^{g,k+1}=-\Delta\lambda_{ij,t}^{g,k+1},$$
$$\Delta p_{ij,t}^{g,k+1}=\Pi_{p^{\max}}\left(\frac{\Delta p_{ij,t}^{i,k+1}-\Delta p_{ji,t}^{j,k+1}}{2}+\frac{y_{ij,t}^{p,k}-y_{ji,t}^{p,k}}{2\rho}\right),$$
$$\Delta p_{ji,t}^{g,k+1}=-\Delta p_{ij,t}^{g,k+1},\quad \forall i<j,t,$$

where $\Pi_c(x)=\min\{\max\{x,-c\},c\}$. The diagonal elements $\Delta\lambda_{ii,t}^{g,k+1}$ and $\Delta p_{ii,t}^{g,k+1}$ are fixed at zero. The parameter $\lambda_{ij}^{\max}$ is the workload-transfer limit of link (i,j), and $p^{\max}$ is the inter-VPP energy-transfer limit.

The global purchased-power variables and d are updated by solving

$$\begin{aligned}&\left(\{\bar{p}_{i,t}^{\mathrm{b},g,k+1}\}_{i,t},d^{k+1}\right)\\&=\underset{\{\bar{p}_{i,t}^{\mathrm{b},g}\}_{i,t},d}{\arg\min}\Bigg\{-\sum_{i\in\mathcal{N}}\rho_o(1-d)P_i^D\\&\quad+\sum_{i,t}y_{i,t}^{b,k}\left(\bar{p}_{i,t}^{\mathrm{b},i,k+1}-\bar{p}_{i,t}^{\mathrm{b},g}\right)\\&\quad+\frac{\rho}{2}\sum_{i,t}\left(\bar{p}_{i,t}^{\mathrm{b},i,k+1}-\bar{p}_{i,t}^{\mathrm{b},g}\right)^2\Bigg\}\end{aligned}$$

subject to

$$\left\|\bar{\boldsymbol{L}}-\boldsymbol{L}^{\mathrm{CDL}}\right\|_2\le d,$$
$$\bar{L}_t=\frac{\sum_{i\in\mathcal{N}}\bar{p}_{i,t}^{\mathrm{b},g}}{\sum_{i\in\mathcal{N}}P_i^D},\quad\forall t.$$

Third, the Lagrangian multipliers are updated according to the consistency-constraint violations:

$$y_{ij,t}^{\lambda,k+1}=y_{ij,t}^{\lambda,k}+\rho\left(\Delta\lambda_{ij,t}^{i,k+1}-\Delta\lambda_{ij,t}^{g,k+1}\right),$$
$$y_{ij,t}^{p,k+1}=y_{ij,t}^{p,k}+\rho\left(\Delta p_{ij,t}^{i,k+1}-\Delta p_{ij,t}^{g,k+1}\right),$$
$$y_{i,t}^{b,k+1}=y_{i,t}^{b,k}+\rho\left(\bar{p}_{i,t}^{\mathrm{b},i,k+1}-\bar{p}_{i,t}^{\mathrm{b},g,k+1}\right).$$

To distinguish the stopping tolerance from the BESS self-discharge coefficient ϵ and the server-count variable $s_{i,t}$, it is denoted by ϵ_{ADMM}. Because the consensus variables have different physical scales, their residual components are scaled by the corresponding upper bounds before root-mean-square aggregation. Excluding the diagonal migration elements, the number of consensus components is

$$M_{\mathrm{c}}=2N(N-1)T+NT.$$

The normalized primal residual is calculated as

$$\begin{aligned}r_{\mathrm{pri}}^{k+1}=\Bigg\{\frac{1}{M_{\mathrm{c}}}\Bigg[&\sum_{\substack{i,j\in\mathcal{N}\\i\neq j}}\sum_{t\in\mathcal{T}}\left(\frac{\Delta\lambda_{ij,t}^{i,k+1}-\Delta\lambda_{ij,t}^{g,k+1}}{\lambda_{ij}^{\max}}\right)^2\\&+\sum_{\substack{i,j\in\mathcal{N}\\i\neq j}}\sum_{t\in\mathcal{T}}\left(\frac{\Delta p_{ij,t}^{i,k+1}-\Delta p_{ij,t}^{g,k+1}}{p^{\max}}\right)^2\\&+\sum_{i\in\mathcal{N}}\sum_{t\in\mathcal{T}}\left(\frac{\bar{p}_{i,t}^{\mathrm{b},i,k+1}-\bar{p}_{i,t}^{\mathrm{b},g,k+1}}{P_i^{\mathrm{grid,max}}}\right)^2\Bigg]\Bigg\}^{1/2},\end{aligned}\tag{65}$$

and the normalized dual residual is

$$\begin{aligned}r_{\mathrm{dual}}^{k+1}=\rho\Bigg\{\frac{1}{M_{\mathrm{c}}}\Bigg[&\sum_{\substack{i,j\in\mathcal{N}\\i\neq j}}\sum_{t\in\mathcal{T}}\left(\frac{\Delta\lambda_{ij,t}^{g,k+1}-\Delta\lambda_{ij,t}^{g,k}}{\lambda_{ij}^{\max}}\right)^2\\&+\sum_{\substack{i,j\in\mathcal{N}\\i\neq j}}\sum_{t\in\mathcal{T}}\left(\frac{\Delta p_{ij,t}^{g,k+1}-\Delta p_{ij,t}^{g,k}}{p^{\max}}\right)^2\\&+\sum_{i\in\mathcal{N}}\sum_{t\in\mathcal{T}}\left(\frac{\bar{p}_{i,t}^{\mathrm{b},g,k+1}-\bar{p}_{i,t}^{\mathrm{b},g,k}}{P_i^{\mathrm{grid,max}}}\right)^2\Bigg]\Bigg\}^{1/2}.\end{aligned}\tag{66}$$

The root-mean-square scaling makes the residual components comparable across variable groups and reduces their direct dependence on the number of VPPs. The distributed iterations terminate when $r_{\mathrm{pri}}^k<\epsilon_{\mathrm{ADMM}}$ and $r_{\mathrm{dual}}^k<\epsilon_{\mathrm{ADMM}}$, or when the maximum number of iterations is reached. These criteria assess numerical consensus convergence of the distributed iterations rather than global optimality of the relaxed cooperative problem.

2) Computational Framework and Complexity: The proposed framework is designed for computational scalability. In Phase 1, each ADMM pass exhibits a per-iteration wall-clock complexity of approximately $O((NT)^p+N^2T)$, with $p\in[2,3]$, primarily dictated by solving N local subproblems in parallel. The integer feasibility-recovery pass uses the same decomposition with fixed server counts and therefore does not change the asymptotic complexity order. In Phase 2, the proportional settlement rule requires only $N+1$ coalition evaluations, rather than the exponential number required by the standard Shapley value. Each coalition evaluation comprises a continuous-relaxation solve followed by integer feasibility recovery.

The complete solution procedure is summarized in Algorithm 1. Phase 1 solves the continuous relaxation of the cooperative problem and then recovers and verifies a solution with integer server counts. Phase 2 applies the same relaxation-and-recovery procedure to the independent problems and uses the resulting standalone and grand-coalition costs for proportional settlement.

V. Simulation Experiments and Result Analysis

This section demonstrates the implementation principle of the proposed model and validates its feasibility through practical case studies.

A. Experiment Settings

We consider four of Google's DCs in the United States as representative study subjects, using real electricity prices and weather data from each location [41], [42]. The DC locations are: (1) Council Bluffs, IA, (2) Berkeley County, SC, (3) The Dalles, OR, (4) Lenoir, NC. The setting of distance matrix D is shown in Table I. The straight-line distance matrix is used to characterize the basic geographical separation among the selected DC locations and to construct a simplified proxy for cross-regional migration cost. In addition, identical initial workload requests are adopted in the baseline setting in order

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Algorithm 1 ADMM-Based Distributed Optimization and Proportional Settlement

Input: System parameters, β, γ, ρ, ϵ_{ADMM}, and $K_{\max}$
Phase 1: ADMM Optimization (P2)
Relax $s_{i,t}$; initialize all ADMM variables and multipliers; set $r^0_{\mathrm{pri}} = r^0_{\mathrm{dual}} = +\infty$ and $k = 0$
while $k < K_{\max}$ **and** ($r^k_{\mathrm{pri}} \geq \epsilon_{\mathrm{ADMM}}$ **or** $r^k_{\mathrm{dual}} \geq \epsilon_{\mathrm{ADMM}}$) **do**
  Each VPP$_i$ (parallel): Solve its local continuous subproblem
  Update the global migration and purchased-power variables
  Update the Lagrangian multipliers
  Calculate the residuals using (65) and (66); set $k = k+1$
end while
Recover $\hat{s}_{i,t} = \lceil s^*_{i,t} \rceil$
Fix $s_{i,t} = \hat{s}_{i,t}$, reinitialize the ADMM variables, and repeat the preceding ADMM iterations until the same stopping condition is satisfied
Verify all original constraints and calculate $c(\mathcal{N})$
Phase 2: Proportional Settlement (P3)
for each VPP$_i$, $i \in \mathcal{N}$ **do**
  Solve the continuous relaxation of **P1**, recover $\hat{s}_{i,t}$, and re-optimize with fixed server counts
  Verify all original constraints and obtain $c(\{i\})$
end for
Calculate $V^{\mathrm{save}} = \sum_{i\in\mathcal{N}} c(\{i\}) - c(\mathcal{N})$ and $\theta_i = c(\{i\}) / \sum_{j\in\mathcal{N}} c(\{j\})$
Calculate $C^*_i = c(\{i\}) - \theta_i(1-\gamma)V^{\mathrm{save}}$
Output: $\bar{\boldsymbol{X}}^*$, $c(\{i\})$, $c(\mathcal{N})$, θ_i, and C^*_i

TABLE I
STRAIGHT LINE DISTANCE BETWEEN THE FOUR GEOGRAPHICAL LOCATIONS (UNIT: km)

Location	Index	1	2	3	4
IA	1	0	2048	2540	1764
SC	2	2048	0	4515	423
OR	3	2540	4515	0	4232
NC	4	1764	423	4232	0

to isolate the effect of the proposed coordination strategy under comparable starting conditions.

Fig. 2 shows the fuzzy sets of PV generation data for four geographic locations during the scheduling period. Fig. 3 shows the distribution of interactive workload requests during the scheduling period. In the baseline case, the initial requests of the four DCs are set to be identical to isolate the effect of the proposed coordination strategy under comparable starting conditions, where (a, b, c) denote the triplets of the triangular fuzzy set.

To isolate the contributions of workload migration, energy sharing, and uncertainty handling, four operating modes are considered. In the independent mode, both inter-VPP workload migration and energy sharing are disabled. In the workload-only mode, energy sharing is disabled by setting $p^{\max} = 0$, while workload migration remains available. In the robust coupled mode, both migration mechanisms are retained, while the uncertain workload and PV output are fixed at their

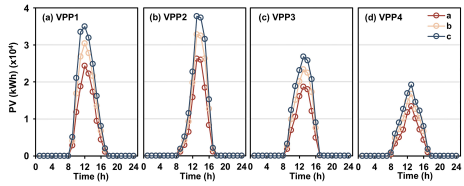


Fig. 2. PV power generation in different geographical locations.

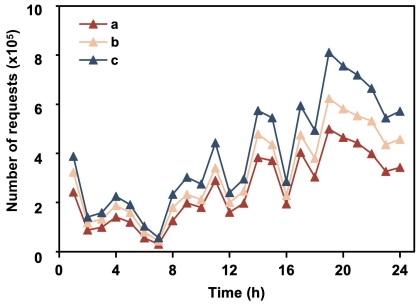


Fig. 3. DC interactive workload requests.

adverse support bounds, i.e., $\lambda_{i,t} = c^{\lambda}_{i,t}$ and $p^{\mathrm{PV}}_{i,t} = a^{\mathrm{PV}}_{i,t}$. The proposed coupled mode jointly coordinates workload migration and energy sharing under the credibility-based fuzzy chance constraints. All modes use the same electricity prices, resource capacities, initial BESS states, CDL trajectory, and solver settings to ensure a consistent comparison.

B. Result Analysis

The performance of the proposed algorithms is validated through convergence and scalability analyses, as shown in Fig. 4 and Fig. 5. Fig. 4 plots the normalized primal and dual residuals of the ADMM iterations. Although moderate fluctuations appear in the early iterations, both residuals decay rapidly and approach zero after approximately 50 iterations, indicating that the consistency violations among decomposed subproblems are effectively eliminated. These results confirm the stable convergence behavior of the proposed ADMM-based distributed optimization framework. Beyond convergence for a fixed-size problem, Fig. 5 investigates the computational scalability of the framework as the number of VPPs (N) increases. As shown in Fig. 5(a), while the centralized solver is faster for small systems, its computation time grows at a super-linear rate. In contrast, the proposed ADMM algorithm exhibits a much flatter, near-linear growth trend, confirming its superior scalability for the optimization phase. This advantage is even more pronounced in the benefit allocation phase, as shown in Fig. 5(b). The traditional Shapley value requires exhaustive coalition evaluation, so its runtime grows exponentially and quickly becomes intractable. By contrast, the proposed settlement rule depends only on the singleton evaluations and the grand-coalition evaluation, and therefore exhibits near-linear computational growth. Together, these results validate that the proposed framework is not only convergent but also computationally scalable, making it suitable for practical applications in large-scale multi-VPP systems.

Table II reports the sensitivity of the relaxation-and-correction strategy to different server scales. The relative gap

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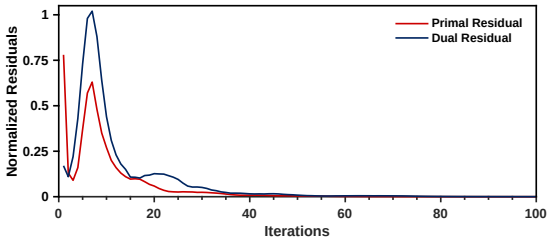
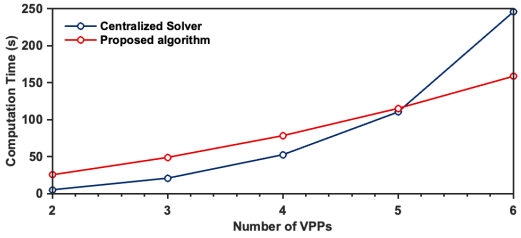
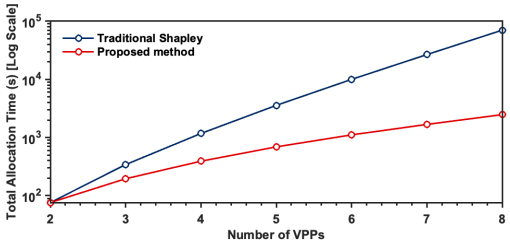


Fig. 4. Normalized primal and dual residuals of the ADMM iterations.



(a) Computational performance of the optimization algorithm.



(b) Computational performance of the benefit allocation method.

Fig. 5. Scalability comparison of the proposed optimization and allocation algorithms with respect to the number of VPPs.

TABLE II
SENSITIVITY ANALYSIS OF THE RELAXATION STRATEGY

Server scale	Relaxed objective ($\times 10^4$\$)	Corrected objective ($\times 10^4$\$)	Relative gap (%)
50	1.2486	1.2732	1.97
100	2.4914	2.5192	1.12
200	4.9865	5.0174	0.62
500	12.4673	12.4982	0.25
1000	24.9359	24.9740	0.15
5000	124.7516	124.8523	0.08
10000	249.5105	249.6351	0.05

decreases rapidly as the server scale increases, indicating that the discretization effect is more pronounced in small-scale cases but becomes negligible for medium- and large-scale DCs. This observation is consistent with the theoretical analysis in Section IV-A. For small-scale scenarios, exact mixed-integer solvers can be adopted when higher integer precision is required.

Fig. 6 illustrates the sensitivity of the scheduling results to the confidence level β. As β increases, the total operating cost rises gradually, while the average server utilization decreases slightly. According to the deterministic equivalents derived in Section III-C, a higher β increases the confidence-adjusted workload bound and decreases the available PV bound, thereby tightening the server-capacity and power-balance constraints. The results therefore show the trade-off between operating cost

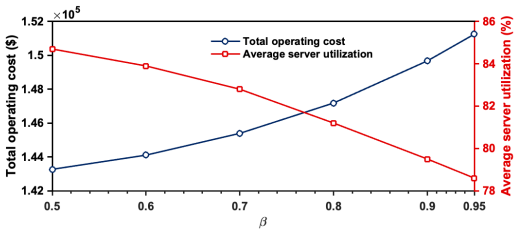


Fig. 6. Sensitivity analysis of the confidence level parameter.

TABLE III
SENSITIVITY ANALYSIS UNDER DIFFERENT HISTORICAL WORKLOAD SAMPLE SIZES

Historical sample ratio	Total operating cost ($\times 10^5$\$)	Average server utilization (%)	Euclidean distance
100%	1.4657	73.84	0.0630
75%	1.4689	73.52	0.0641
50%	1.4728	73.07	0.0654
25%	1.4786	72.41	0.0672
10%	1.4869	71.68	0.0695

and scheduling conservativeness controlled by the confidence level.

Table III compares the scheduling results under different historical workload sample sizes used to construct the triangular fuzzy parameters. When the historical sample ratio decreases from 100% to 10%, the total operating cost increases from 1.4657×10^5 \$ to 1.4869×10^5 \$, while the average server utilization decreases from 73.84% to 71.68% and the Euclidean distance increases from 0.0630 to 0.0695. These results show a gradual degradation as workload information becomes more limited. Since the fuzzy formulation requires only the lower, nominal, and upper estimates rather than an accurately estimated probability distribution, the scheduling model can still be constructed under limited historical workload data.

Fig. 7 and Table IV compare the CDL-tracking performance under independent and cooperative operation. Under independent operation, relatively large deviations occur during peak hours (16–22 h), with VPP_2 exhibiting the largest Euclidean distance ($d_2 = 0.120$) and the lowest similarity ($\varepsilon_2 = 0.880$), while VPP_1 achieves the smallest deviation ($d_1 = 0.066$, $\varepsilon_1 = 0.934$). Under cooperative operation, the aggregate response achieves a Euclidean distance of $d = 0.063$ and a similarity of $\varepsilon = 0.937$, with the corresponding DR incentive increasing from 51,753.60 dollars to 53,962.79 dollars. The cooperative profile represents the aggregate grid-interface response resulting from local DC, PV, and BESS scheduling together with inter-VPP workload and energy coordination. The results therefore show that coordinating computing- and energy-side flexibility at the VPP level provides greater flexibility for aggregate CDL tracking.

Table V and Fig. 8 illustrate the contribution of the energy-sharing mechanism. Compared with workload-only migration, the proposed coupled mode reduces the peak single-link workload migration from 425.6×10^3 to approximately 220.0×10^3 requests and the maximum link-capacity utilization from 94.5% to approximately 48.9%. The total workload migration decreases from 3150.8×10^3 to 1845.2×10^3 requests,

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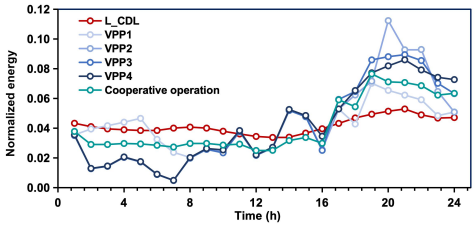


Fig. 7. Normalized DR curves during the scheduling period.

TABLE IV
COMPARISON OF THE EFFECTIVENESS OF DIFFERENT OPERATING MODELS FOR PARTICIPATING IN DR

Mode of operation	Independent operation				Cooperative operation
DR parameters	VPP1	VPP2	VPP3	VPP4	
Euclidean distance (d_i, d)	0.066	0.120	0.113	0.107	0.063
similarity $(\varepsilon_i, \varepsilon)$	0.934	0.880	0.887	0.893	0.937
Total DR incentive ($)	51753.60				53962.79

TABLE V
COMPARISON OF CROSS-REGIONAL WORKLOAD-TRANSFER PRESSURE UNDER DIFFERENT COORDINATION MODES

Metric	Independent	Workload-only	Coupled
Peak single-link migration (10^3)	0	425.6	220.0
Max capacity utilization (%)	0	94.5	48.9
Total task migration (10^3)	0	3150.8	1845.2
Total energy transfer (MWh)	0	0	142.6

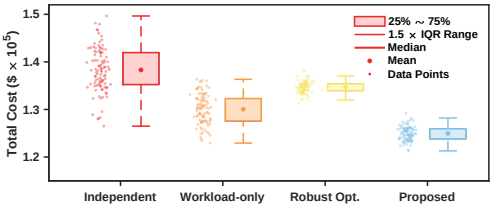


Fig. 8. Comparison of total cost distributions under different operation modes.

corresponding to a reduction of approximately 41.4%. Meanwhile, 142.6 MWh of inter-VPP energy transfer is scheduled as an additional spatial adjustment channel.

As shown in Fig. 8, workload-only migration reduces the operating cost relative to independent operation, whereas the proposed coupled mode achieves the lowest median cost among the compared modes. The robust coupled mode incurs a higher operating cost because its scheduling decisions are based on the adverse workload and PV bounds. These results indicate that energy sharing complements workload migration by reducing cross-regional workload-transfer pressure while maintaining favorable economic performance.

Fig. 9 reveals that all BESS units across the four VPPs exhibit synchronized operation patterns despite geographical separation, with charging predominantly during midday (11-16h) and discharging during evening price peaks (18-22h), and their operation shows remarkable consistency, suggesting electricity price signals drive BESS dispatch decisions. This coordinated BESS operation provides crucial temporal flexibility.

Fig. 10 illustrates the signed bilateral workload transfers

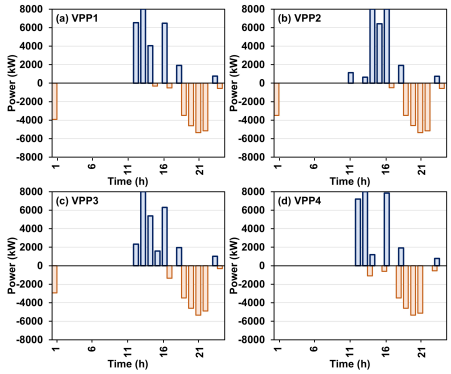


Fig. 9. BESS charging/discharging schedule decisions.

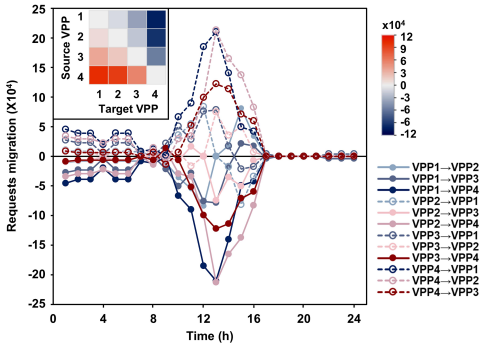


Fig. 10. Interactive workload migration volumes among geo-distributed VPPs.

among the VPPs under the proposed coupled mode. The heatmap reports the cumulative net transfer for each VPP pair, with VPP_4 acting as the main workload exporter and VPP_1 and VPP_2 as the principal receivers. Workload migration is mainly concentrated during 10–16 h, with the peak absolute single-link transfer reaching approximately 22×10^4 requests at around 13 h. Moderate transfers occur during 0–8 h, while the transfer volumes become negligible after 16 h.

The pairwise trajectories are equal in magnitude and opposite in sign, showing that the computed schedule satisfies the antisymmetric workload-transfer constraints in (18). These results indicate that workload migration redistributes computing demand across locations during the main coordination period while preserving system-wide workload conservation.

Fig. 11 presents the signed bilateral energy-sharing schedules among the VPPs. The heatmap reports the cumulative net energy transfer for each VPP pair over the scheduling horizon. In cumulative terms, the VPP_1–VPP_4 link exhibits the largest transfer, with approximately 6×10^3 kWh transferred from VPP_1 to VPP_4 and an equal-magnitude opposite entry for the reverse direction. In terms of the time-dependent trajectories, the largest absolute single-period transfer occurs on the VPP_2–VPP_4 link, reaching approximately 1.1×10^3 kWh at around 13 h. Energy-sharing activities are most pronounced during 11–16 h, while their magnitudes are relatively small outside this interval.

The predominant energy-sharing direction toward VPP_4 is opposite to the main workload-migration direction from VPP_4 toward VPP_1 and VPP_2 shown in Fig. 10. Together with the reduced workload-transfer requirement reported in Table V,

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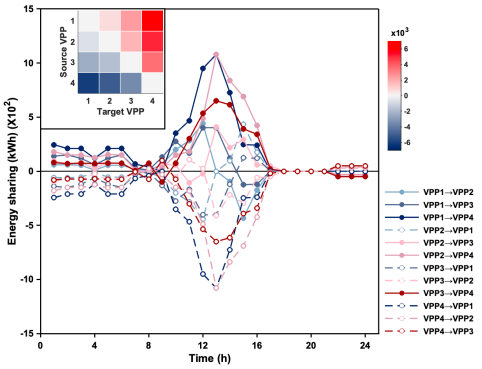


Fig. 11. Energy-sharing volumes among geo-distributed VPPs.

TABLE VI
COMPARISON OF OPERATION COSTS FOR DIFFERENT VPPS

Mode of operation	VPP1	VPP2	VPP3	VPP4	SUM
Independent operation ($)	47432.3	42628.4	27244.2	39644.1	156949.0
Cooperative operation ($)	43611.3	41717.0	25904.5	35338.7	146571.5

this opposite directional pattern indicates that energy sharing provides a complementary spatial adjustment channel, thereby reducing the reliance on additional cross-regional workload migration in the coupled scheduling process.

Table VI demonstrates the economic advantages of cooperative operation across all VPPs. Under independent operation, the total system cost reaches 156,949.0 dollars, while cooperative operation reduces this to 146,571.5 dollars, yielding significant overall savings of 10,377.5 dollars (6.6%). This confirms that the proposed bidirectional migration strategy effectively leverages geographical complementarity to reduce system-wide operational costs while maintaining individual economic incentives for all participants.

VI. CONCLUSION

This paper proposed an energy-workload coupled migration optimization strategy for VPPs with DCs. A coordinated migration model based on antisymmetric matrices was established, and fuzzy chance-constrained modeling was introduced for workload and photovoltaic uncertainty. The corresponding SOCP reformulation was further derived to address uncertainty under sparse workload data. In addition, a scalable coordination and settlement scheme was developed for multi-VPP cooperation, in which a distributed ADMM algorithm was used for large-scale coupled scheduling and a proportional settlement rule was adopted to avoid exhaustive coalition evaluation in exact Shapley settlement. Simulations using real data from Google's DCs validated the effectiveness of the proposed method in reducing operational costs while enhancing DR curve tracking precision. Future research could explore adaptive tuning of penalty parameters and migration strategies to cope with real-time system dynamics and market volatility.

APPENDIX A
ANALYTICAL PROPERTIES OF THE PROPOSED ALLOCATION MECHANISM

This appendix establishes the individual rationality and settlement balance of the proposed allocation rule under non-negative grand-coalition savings.

Assume that the grand coalition is weakly beneficial, that is

$$V^{\text{save}} = \sum_{i\in\mathcal{N}} c(\{i\}) - c(\mathcal{N}) \geq 0. \tag{67}$$

Provided that $c(\{i\}) \geq 0$ for all $i \in \mathcal{N}$ and $\sum_{i\in\mathcal{N}} c(\{i\}) > 0$, the proportional weights defined in (32) satisfy:

$$\theta_i = \frac{c(\{i\})}{\sum_{j\in\mathcal{N}} c(\{j\})} \geq 0, \qquad \forall i \in \mathcal{N},$$
$$\sum_{i\in\mathcal{N}} \theta_i = 1. \tag{68}$$

This direct definition remains valid when $V^{\text{save}} = 0$.

According to (33),

$$c(\{i\}) - C_i^* = \theta_i(1-\gamma)V^{\text{save}} \geq 0, \qquad \forall i \in \mathcal{N}, \tag{69}$$

where $\gamma \in [0, 1)$. Therefore

$$C_i^* \leq c(\{i\}), \qquad \forall i \in \mathcal{N}, \tag{70}$$

which establishes individual rationality because no VPP incurs a higher settled cost than under independent operation.

Summing (33) over all VPPs and using (67), (68), and (28) gives

$$\begin{aligned}\sum_{i\in\mathcal{N}} C_i^* &= \sum_{i\in\mathcal{N}} c(\{i\}) - (1-\gamma)V^{\text{save}} \sum_{i\in\mathcal{N}} \theta_i \\ &= c(\mathcal{N}) + \gamma V^{\text{save}} \\ &= c(\mathcal{N}) + V^{\text{O}}.\end{aligned} \tag{71}$$

Thus, the total settlement covers the grand-coalition operating cost and the VPPO service fee. The VPPs retain $(1-\gamma)V^{\text{save}}$, while the VPPO receives γV^{save}.

The proposed rule therefore ensures individual rationality and settlement balance under the grand coalition. It does not establish core stability, which would require evaluating all nonempty sub-coalitions and entail $O(2^N)$ coalition-cost evaluations. The proposed mechanism instead requires only $N + 1$ evaluations.

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